Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/SQL validation pipelines**, **automation tooling**, and **quality monitoring workflows** for large-scale AI systems, with strong testing, analytics, and release reliability focus.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **automated Python/FastAPI pipelines** for 5K+ monthly jobs with validation checks, error handling, and structured metadata outputs, enabling consistent quality assessment and reliable analysis across production data workflows.
- Designed SQL-driven monitoring paths for 500+ users, reducing latency from **1.2s to 350ms** while improving signal quality for dashboard metrics, anomaly detection, and release decision support processes.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Developed **behavior analysis pipelines** over 100K+ trajectory logs to benchmark model decisions, compare scenario outcomes, and generate reproducible summaries for rapid validation and iterative refinement workflows.
- Created technical diagnostics and experiment reports used by researchers and engineers, improving communication of validation conclusions and accelerating conversion of prototype analyses into production-ready methodologies.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Implemented **high-scale automation workflows** for 10K+ daily requests, integrating OCR/model outputs with robust test coverage to maintain stable validation behavior under changing requirements and large dataset volumes.
- Used AI-assisted coding workflows and CI/CD quality gates to reduce turnaround by **60%**, improving maintainability and reliability of analysis pipelines deployed in fast-moving product release cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **data processing and ETL services** with Python/Java, AWS Glue, and S3, improving response times by **30%** while enabling scalable validation of transformed datasets across distributed systems.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, boosting CI pipeline efficiency by **68%** and reducing defect escape risk through continuous automated verification practices.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built real-time REST/WebSocket modules supporting 2M+ synchronized records and eight workflows, improving consistency of event-driven behavior and traceability of operational state transitions during runtime validation.
- Automated Docker/Jenkins deployment and test stages, reducing release time by **50%** while improving repeatability of build artifacts and reliability of ongoing integration and maintenance workflows.

Skills

Core: **Python**, **SQL**, Java, JavaScript/TypeScript, C++
Validation & Analytics: behavior metrics, large-dataset analysis, automated checks, quality reporting, scenario benchmarking
Engineering: FastAPI, Flask, REST APIs, CI/CD, Git, Jenkins, GitHub Actions, unit/integration testing
Data/Platform: PostgreSQL, MySQL, AWS (S3/Glue), Azure SQL, Docker, Kubernetes, observability-minded debugging

Projects

CaptionCraft - Image Caption Generator | *Python, Transformers, Vision Models*

- Benchmarked BLIP, GIT, and ViT-GPT2 on a 10K-image and 100K-caption dataset, improving caption accuracy from **81% to 96%** through reproducible evaluation workflows and model-level validation metrics.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a schema-aware query analysis workspace for 10+ databases with structured validation and execution summaries, improving reliability of large-scale data exploration and accelerating investigation of anomalous query behavior.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Implemented a containerized data API with automated validation checks and CI/CD deployment controls, achieving **87%** processing accuracy while improving consistency and traceability of production analysis outputs.